

Mingtian Gong

(+1) 9708880240

emoyoqaq@gmail.com

EDUCATION

NYU Tandon school of engineering

2022.08 – 2024.05

Master of Science in Computer Engineering

- **Coursework (Computer Science):** PRNCPLS DATABASE SYSTEM, INTRO TO OPERATING SYSTEM
- **Coursework (Computer Engineering):** Data structure Algorithm, Computing System Architecture, Probability and Stochastic, Internet Architecture & Protocol, Intro to Operating System, Real Time Embedded System Design, Machine Learning, Intro to Wireless Networking

Colorado State University

2018.09 - 2021.12

Bachelor of Science in Computer Science

- **Coursework (Math):** Linear Algebra, Intro to Combinatorial Theory, Intro to Mathematical Modeling
- **Coursework (Computer Science):** Intro to Programming (CS0) - Java, Prior Programming Experience, Data Structure, Discrete Structures and their applications, Software Development with C++, Computer Organization, Software Engineering, Algorithms-Theory and Practice, System Security, Operating Systems, Database System, Object-Oriented Design, Intro to Big Data, Intro to Artificial Intelligence

RESEARCH EXPERIENCES

Machine Learning and Computer Vision and Applications to Self-Driving Car

2021.10 - 2021.12

Researcher | Advisor: Dr. Ian Deng UCSD

- Implemented advanced filtering algorithms, non-linear filtering, and patches
- Implemented algorithm for image feature extraction, such as Edge Detection, line detection, Corner detection, 3D stereo vision
- Experimented on object recognition and image understanding
- Applied statistical learning theory and pattern recognition algorithm for image segmentation, such as Bayesian decision theory, maximum likelihood
- Designed self-driving car central control system, including lane detection, traffic sign classification, advanced lane finding, vehicle detection, and tracking, road segmentation

COURSE PROJECTS

Online Chess Platform development

2021.8 - 2021.12

- Designed GUI using **CSS**, **HTML** and added different features with **React**
- Used npm to install external source of default chess board
- Practiced **Agile** methodologies during the semester in total of 3 sprints, worked as a sprint master of one sprint, hosted daily standup, setup sprint goal and so on.
- Created invitation functions to allow players to invite their friends to play against each other
- Implemented rules of movement of each chess piece
- Managed interactions among the client, the server, and the database
- Employed **MariaDB** to store user registration information, matching history, and game state.
- Wrote **Unit Tests** to ensure code coverage higher than 80%

Mapping Ideal Traffic Routes Using The open Street Maps Dataset

2021.8 - 2021.12

- Used OSMnx python package (OSMNX 1.1.2) to extract the needed sub-dataset from the PlanetOSM dataset from Open Street Maps whose size is about 110 GB
- Wrote a Spark program to process the sub-dataset in Scala
- Designed an adapted version of Dijkstra's shortest path algorithm to store the actual paths between nodes rather than just distances
- Converted the above algorithm to one that uses Pregels because it's almost impossible to use Dijkstra's shortest path algorithm directly on such a massive dataset.
- Wrote a program to return a CSV file containing the node identification number, latitude, and longitude for every point in the path and plot it to compare with other Mapping programs such as Google map.

Interactive map development

2021.1 - 2021.5

- Implemented and tested an interactive map using **React** and **Java**

- Practiced **Agile** methodologies during the semester in total of 4 sprints, worked as a sprint master of one sprint, hosted daily standup, setup sprint goal and so on.
- Designed interactions between the server and databases using **MVC** pattern
- Implemented frontend functions using **React** and backend functions using **Java** to add features to the map
- Wrote **Unit test** to keep code coverage over 80%

CS 435 Introduction to Big Data Course Project

2021.10 - 2021.11

- Designed, implemented, and tested a Spark program in Scala
- Generated a sorted list of Wikipedia pages based on their ideal PageRank value in descending order
- Generated a sorted list (in descending order) of Wikipedia pages based on their PageRank value with taxation
- Generated a Wikipedia Bomb that returns the “Rocky Mountain National Park” Wikipedia page for the search key word “surfing”

CS 435 Introduction to Big Data Course Project

2021.09 - 2021.10

- Designed, implemented, and tested a program with Java Hadoop Framework
- Used big data files to simulate small-world friend networks as input files
- Calculated geodesic Path between networks
- Calculated the clustering coefficient: large degree of transitivity, when the path length is 1 to 8

CS 370 Operating System Course Project

2021.04 - 2021.05

- Designed, implemented, and tested a program in Java that used multi-threading to create many producers and consumers: the producer generated random numbers and added them to the buffer; The consumer consumed the random numbers in the buffer. Producers and consumers accessed the same buffer through synchronization.

CS 370 Operating System Course Project

2021.03 - 2021.04

- Designed, implemented, and tested a program in Python to simulate three CPU scheduling algorithms, such as SJF(Shortest Job First), FCFS(First Come First Serve), and RR(Round Robin)

CS 356 System Securities Course Project

2021.02 - 2021.03

- Designed, implemented, and tested a program for file encryption and decryption in C++

CS 370 Operating System Course Project

2021.02 - 2021.03

- Designed, implemented, and tested a program in C, which uses a fork to create three child processes that output the first n Fibonacci numbers, the first n primes, and the sum of the first n numbers, respectively
- Utilized Pipe and shared memory

CS 370 Operating System Course Project

2021.01 - 2021.02

- Designed a program that randomly and dynamically allocates memory for arrays
- Used Valgrind framework to check if there is a memory leak

CS 253 Software Development with C++ Course Project

2020.08 - 2020.12

- Simulated Conway's Game of Life and designed a game that users can create an initial configuration and observe how it evolves
- Implemented the program to allow players to customize the appearance of cells, the rules of cell Life and death, and the rate of cell iteration through animation

CS 253 Software Development with C++ Course Project

2020.08 - 2020.12

- Wrote a Tokenizer Program; tokenized the input files, and continued to improve the program

SKILLS

Languages: Java, C, C++, Python, MATLAB, JavaScript, CSS, HTML, SQL

Tools & Frameworks: IntelliJ, Eclipse, Visual Studio Code, Docker, DETERLab, MobaXterm, Spring, Springboot, REST API, CI/CD, Hibernate, Microservices, Junit5, React, MariaDB, Valgrind, Hadoop MapReduce, Agile, Apache Kafka, Jenkins, Git, Linux